TASK B: Online Shopping Product Management System

Develop an ASP.NET Core MVC application to display products, calculate discounted prices and manage customer selections.

1. Model and Extension Method

o Create a Product model with ProductId, ProductName, Category, Price and DiscountPercentage.

o Implement GetDiscountedPrice() to calculate the selling price.

o Display product details and discounted price.

2. Asynchronous Web Request

o Retrieve product-related sample data asynchronously from https://fakestoreapi.com/products.

o Display product title, price and category in a table.

3. State Management

o Cookie: Store customer name.

o Session: Store selected product category.

o Query String: Pass ProductId to the product details page.

o Hidden Field: Submit ProductId with an order form.

4. Caching

o Implement Response Caching for the product details page.

o Cache the product list using IMemoryCache with a 10-minute expiration.

NOTE: Students must write appropriate comments for each line of code to explain its purpose and functionality. Comments should clearly describe the logic, methods, variables and important statements used in the program.

Product.cs

```
namespace ProductManagementSystem.Models
{
    public class Product
    {
        public int ProductId { get; set; }

        public string ProductName { get; set; } = string.Empty;

        public string Category { get; set; } = string.Empty;

        public decimal Price { get; set; }

        public decimal DiscountPercentage { get; set; }

        public decimal GetDiscountedPrice()
        {
            return Price - (Price * DiscountPercentage / 100);
        }
    }
}
```

ProductApiModel.cs

```
namespace ProductManagementSystem.Models
{
    public class ProductApiModel
    {
        public int Id { get; set; }

        public string Title { get; set; } = string.Empty;

        public decimal Price { get; set; }

        public string Category { get; set; } = string.Empty;
```

```
    }
}
```

ProductApiService.cs

```
namespace ProductManagementSystem.Models
{
    public class ProductApiModel
    {
        public int Id { get; set; }

        public string Title { get; set; } = string.Empty;

        public decimal Price { get; set; }

        public string Category { get; set; } = string.Empty;
    }
}
```

OrderViewModel.cs

```
namespace ProductManagementSystem.Models
{
    public class OrderViewModel
    {
        public int ProductId { get; set; }

        public string CustomerName { get; set; } = string.Empty;

        public int Quantity { get; set; }
    }
}
```

Program.cs

```
using ProductManagementSystem.Services;

var builder = WebApplication.CreateBuilder(args);

// MVC
builder.Services.AddControllersWithViews();

// Session
builder.Services.AddDistributedMemoryCache();

builder.Services.AddSession(options =>
{
    options.IdleTimeout = TimeSpan.FromMinutes(30);
    options.Cookie.HttpOnly = true;
    options.Cookie.IsEssential = true;
});

// HttpClient
builder.Services.AddHttpClient<ProductApiService>();

// Memory cache
builder.Services.AddMemoryCache();

// Response caching
builder.Services.AddResponseCaching();

var app = builder.Build();

if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    app.UseHsts();
}

app.UseHttpsRedirection();

app.UseStaticFiles();

app.UseRouting();

// Response caching middleware
app.UseResponseCaching();
```

```
// Session middleware
app.UseSession();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Products}/{action=Index}/{id?}");

app.Run();
```

Models/ProductaController.cs

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.Extensions.Caching.Memory;
using ProductManagementSystem.Models;
using ProductManagementSystem.Services;

namespace ProductManagementSystem.Controllers
{
    public class ProductsController : Controller
    {
        private readonly ProductApiService _productApiService;
        private readonly IMemoryCache _memoryCache;

        private const string ProductCacheKey = "ProductList";

        public ProductsController(
            ProductApiService productApiService,
            IMemoryCache memoryCache)
        {
            _productApiService = productApiService;
            _memoryCache = memoryCache;
        }

        // Product list
        public async Task<IActionResult> Index()
        {
```

```
// Cookie: store customer name
string? customerName = Request.Cookies["CustomerName"];

if (string.IsNullOrEmpty(customerName))
{
    customerName = "Guest";

    Response.Cookies.Append(
        "CustomerName",
        customerName,
        new CookieOptions
        {
            Expires = DateTimeOffset.UtcNow.AddDays(7),
            HttpOnly = true
        });
}

// IMemoryCache
if (!_memoryCache.TryGetValue(
        ProductCacheKey,
        out List<Product>? products))
{
    products = await _productApiService.GetProductsAsync();

    var cacheOptions = new MemoryCacheEntryOptions
    {
        AbsoluteExpirationRelativeToNow =
            TimeSpan.FromMinutes(10)
    };

    _memoryCache.Set(
        ProductCacheKey,
        products,
        cacheOptions);
}

// Session: selected category
ViewBag.SelectedCategory =
    HttpContext.Session.GetString("SelectedCategory");
```

```
        ViewBag.CustomerName = customerName;

        return View(products);
    }

    // Save selected category in Session
    [HttpPost]
    public IActionResult SetCategory(string category)
    {
        HttpContext.Session.SetString(
            "SelectedCategory",
            category);

        return RedirectToAction(nameof(Index));
    }

    // Product details
    // Query String: ?productId=1
    [ResponseCache(
        Duration = 60,
        Location = ResponseCacheLocation.Any,
        NoStore = false)]
    public async Task<IActionResult> Details(int productId)
    {
        if (productId <= 0)
        {
            return BadRequest();
        }

        if (!_memoryCache.TryGetValue(
                ProductCacheKey,
                out List<Product>? products))
        {
            products = await _productApiService.GetProductsAsync();

            _memoryCache.Set(
                ProductCacheKey,
                products,
                new MemoryCacheEntryOptions
                {
```

```
                    AbsoluteExpirationRelativeToNow =
                        TimeSpan.FromMinutes(10)
                });
        }

        var product = products?
            .FirstOrDefault(p => p.ProductId == productId);

        if (product == null)
        {
            return NotFound();
        }

        return View(product);
    }

    // Display order form
    [HttpGet]
    public IActionResult Order(int productId)
    {
        var model = new OrderViewModel
        {
            ProductId = productId,
            Quantity = 1
        };

        return View(model);
    }

    // Submit order
    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Order(OrderViewModel model)
    {
        if (!ModelState.IsValid)
        {
            return View(model);
        }

        // ProductId comes from hidden field
```

```
            TempData["Message"] =
                $"Order placed successfully for Product ID " +
                $"{model.ProductId}. Quantity: {model.Quantity}.";

            return RedirectToAction(nameof(Index));
        }
    }
}
```

Index.cshtml

```
@model List<ProductManagementSystem.Models.Product>

@{
    ViewData["Title"] = "Products";
}

<h1>Product List</h1>

<p>
    Welcome, <strong>@ViewBag.CustomerName</strong>
</p>

@if (ViewBag.SelectedCategory != null)
{
    <p>
        Selected Category:
        <strong>@ViewBag.SelectedCategory</strong>
    </p>
}

@if (TempData["Message"] != null)
{
    <div class="alert alert-success">
        @TempData["Message"]
    </div>
}
```

```
<h3>Select Category</h3>

<form asp-action="SetCategory" method="post">
   <select name="category" class="form-control mb-2">
       <option value="electronics">Electronics</option>
       <option value="jewelery">Jewellery</option>
       <option value="men's clothing">Men's Clothing</option>
       <option value="women's clothing">Women's Clothing</option>
   </select>

   <button type="submit" class="btn btn-primary">
       Save Category
   </button>
</form>

<hr />

<table class="table table-bordered table-striped">

   <thead>
       <tr>
           <th>Product ID</th>
           <th>Product Name</th>
           <th>Category</th>
           <th>Price</th>
           <th>Discount</th>
           <th>Discounted Price</th>
           <th>Action</th>
       </tr>
   </thead>

   <tbody>

       @foreach (var product in Model)
       {
           <tr>
               <td>@product.ProductId</td>

               <td>
                   @product.ProductName
```

```
            </td>

            <td>
                @product.Category
            </td>

            <td>
                $@product.Price.ToString("0.00")
            </td>

            <td>
                @product.DiscountPercentage%
            </td>

            <td>
                <strong>
                    $@product.GetDiscountedPrice().ToString("0.00")
                </strong>
            </td>

            <td>
                <!-- Query String -->
                <a asp-controller="Products"
                   asp-action="Details"
                   asp-route-productId="@product.ProductId"
                   class="btn btn-info">
                    Details
                </a>

                <a asp-controller="Products"
                   asp-action="Order"
                   asp-route-productId="@product.ProductId"
                   class="btn btn-success">
                    Order
                </a>
            </td>
        </tr>
    }

</tbody>
```

```
</table>
```

Details.cshtml

```
@model ProductManagementSystem.Models.Product

@{
    ViewData["Title"] = "Product Details";
}

<h1>Product Details</h1>

<div class="card">
    <div class="card-body">

        <h3>@Model.ProductName</h3>

        <p>
            <strong>Product ID:</strong>
            @Model.ProductId
        </p>

        <p>
            <strong>Category:</strong>
            @Model.Category
        </p>

        <p>
            <strong>Original Price:</strong>
            $@Model.Price.ToString("0.00")
        </p>

        <p>
            <strong>Discount:</strong>
            @Model.DiscountPercentage%
        </p>

        <p>
```

```
            <strong>Discounted Price:</strong>
            <span class="text-success">
                $@Model.GetDiscountedPrice().ToString("0.00")
            </span>
        </p>

        <a asp-action="Order"
           asp-route-productId="@Model.ProductId"
           class="btn btn-success">
            Place Order
        </a>

        <a asp-action="Index"
           class="btn btn-secondary">
            Back
        </a>

    </div>
</div>
```

Order.cshtml

```
@model ProductManagementSystem.Models.OrderViewModel

@{
    ViewData["Title"] = "Place Order";
}

<h1>Place Order</h1>

<form asp-action="Order" method="post">

    @Html.AntiForgeryToken()

    <!-- Hidden Field -->
    <input type="hidden"
           asp-for="ProductId" />

    <div class="mb-3">
```

```
        <label asp-for="CustomerName"
               class="form-label">
            Customer Name
        </label>

        <input asp-for="CustomerName"
               class="form-control" />
    </div>

    <div class="mb-3">
        <label asp-for="Quantity"
               class="form-label">
            Quantity
        </label>

        <input asp-for="Quantity"
               type="number"
               min="1"
               class="form-control" />
    </div>

    <button type="submit"
            class="btn btn-primary">
        Place Order
    </button>

    <a asp-action="Index"
       class="btn btn-secondary">
        Cancel
    </a>

</form>
```

OUTPUT

ProductManagementSystem Home Privacy

Product List

Welcome, **Guest**

Select Category

Electronics

Save Category

Product ID	Product Name	Category	Price	Discount	Discounted Price	Action
1	Fjallraven - Foldsack No. 1 Backpack, Fits 15 Laptops	men's clothing	$109.95	10%	**$98.96**	Details Order
2	Mens Casual Premium Slim Fit T-Shirts	men's clothing	$22.30	10%	**$20.07**	Details Order
3	Mens Cotton Jacket	men's clothing	$55.99	10%	**$50.39**	Details Order

3	Mens Cotton Jacket	men's clothing	$55.99	10%	**$50.39**	Details Order
4	Mens Casual Slim Fit	men's clothing	$15.99	10%	**$14.39**	Details Order
5	John Hardy Women's Legends Naga Gold & Silver Dragon Station Chain Bracelet	jewelery	$695.00	10%	**$625.50**	Details Order
6	Solid Gold Petite Micropave	jewelery	$168.00	10%	**$151.20**	Details Order
7	White Gold Plated Princess	jewelery	$9.99	10%	**$8.99**	Details Order
8	Pierced Owl Rose Gold Plated Stainless Steel Double	jewelery	$10.99	10%	**$9.89**	Details Order
9	WD 2TB Elements Portable External Hard Drive - USB 3.0	electronics	$64.00	10%	**$57.60**	Details Order
10	SanDisk SSD PLUS 1TB Internal SSD - SATA III 6 Gb/s	electronics	$109.00	10%	**$98.10**	Details Order
11	Silicon Power 256GB SSD 3D NAND A55 SLC Cache Performance Boost SATA III 2.5	electronics	$109.00	10%	**$98.10**	Details Order
12	WD 4TB Gaming Drive Works with Playstation 4 Portable External Hard Drive	electronics	$114.00	10%	**$102.60**	Details Order
13	Acer SB220Q bi 21.5 inches Full HD (1920 x 1080) IPS Ultra-Thin	electronics	$599.00	10%	**$539.10**	Details Order
14	Samsung 49-Inch CHG90 144Hz Curved Gaming Monitor (LC49HG90DMNXZA) – Super Ultrawide Screen QLED	electronics	$999.99	10%	**$899.99**	Details Order
15	BIYLACLESEN Women's 3-in-1 Snowboard Jacket Winter Coats	women's clothing	$56.99	10%	**$51.29**	Details Order
16	Lock and Love Women's Removable Hooded Faux Leather Moto Biker Jacket	women's clothing	$29.95	10%	**$26.96**	Details Order

15	BIYLACLESEN Women's 3-in-1 Snowboard Jacket Winter Coats	women's clothing	$56.99	10%	**$51.29**	Details Order
16	Lock and Love Women's Removable Hooded Faux Leather Moto Biker Jacket	women's clothing	$29.95	10%	**$26.96**	Details Order
17	Rain Jacket Women Windbreaker Striped Climbing Raincoats	women's clothing	$39.99	10%	**$35.99**	Details Order
18	MBJ Women's Solid Short Sleeve Boat Neck V	women's clothing	$9.85	10%	**$8.87**	Details Order
19	Opna Women's Short Sleeve Moisture	women's clothing	$7.95	10%	**$7.16**	Details Order
20	DANVOUY Womens T Shirt Casual Cotton Short	women's clothing	$12.99	10%	**$11.69**	Details Order